

MAT 3150 — Homework 3

Problem 2 Walkthrough: Computing a Limit from the Definition

Goal. Use the limit definition to compute

$$\lim_{n \rightarrow \infty} \frac{n+2}{3n+5}.$$

The definition, in words

$a_n \rightarrow L$ means:

For every $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that $|a_n - L| < \varepsilon$ for all $n \geq N$.

Think of it as a game. Someone hands you a tolerance ε (maybe tiny, like 0.001). You must answer with a point N after which *every* term is within ε of L . A proof is a recipe for N that works no matter which ε you are handed.

Tool you'll need. Archimedean property: for any real number r , there is a natural number N with $N > r$.

The plan (every problem of this type):

1. Guess L .
2. Scratch work: simplify $|a_n - L|$, bound it by something simple, and solve “simple thing $< \varepsilon$ ” for n .
3. Write the proof forward: “Let $\varepsilon > 0$. Choose $N \dots$ ”

Worked example: $\lim_{n \rightarrow \infty} \frac{2n+1}{n+3}$

1. Guess. For large n , the $+1$ and $+3$ barely matter, so $\frac{2n+1}{n+3} \approx \frac{2n}{n} = 2$. Guess $L = 2$.

2. Scratch work. Common denominator:

$$\frac{2n+1}{n+3} - 2 = \frac{(2n+1) - 2(n+3)}{n+3} = \frac{2n+1-2n-6}{n+3} = \frac{-5}{n+3}.$$

The n 's in the numerator cancelled — a good sign the guess is right. So

$$\left| \frac{2n+1}{n+3} - 2 \right| = \frac{5}{n+3}.$$

Simplify by making the denominator *smaller*, which makes the fraction *bigger* (an overestimate is fine):

$$\frac{5}{n+3} < \frac{5}{n}.$$

Now solve $\frac{5}{n} < \varepsilon$ for n : $n > \frac{5}{\varepsilon}$. That tells us how big N must be.

3. The proof (what you hand in).

Proof. We claim the limit is 2. Let $\varepsilon > 0$. By the Archimedean property, choose $N \in \mathbb{N}$ with $N > \frac{5}{\varepsilon}$. Then for all $n \geq N$,

$$\left| \frac{2n+1}{n+3} - 2 \right| = \frac{5}{n+3} < \frac{5}{n} \leq \frac{5}{N} < \varepsilon.$$

Hence $\lim_{n \rightarrow \infty} \frac{2n+1}{n+3} = 2$. □

Why each step in the chain holds:

- $= \frac{5}{n+3}$: the scratch-work algebra.
- $< \frac{5}{n}$: because $n+3 > n$.
- $\leq \frac{5}{N}$: because $n \geq N$.
- $< \varepsilon$: because $N > \frac{5}{\varepsilon}$, i.e. $\frac{5}{N} < \varepsilon$.

Your turn: $\lim_{n \rightarrow \infty} \frac{n+2}{3n+5}$

1. Guess. For large n , $\frac{n+2}{3n+5} \approx \frac{n}{3n} = \underline{\hspace{2cm}}$. So $L = \underline{\hspace{2cm}}$.

2. Scratch work. Put it over a common denominator:

$$\frac{n+2}{3n+5} - L = \frac{\hspace{1cm}}{\hspace{1cm}}.$$

Tip: if L is a fraction, multiply out carefully. If your guess is right, the n 's in the numerator should cancel, leaving a constant.

Take absolute values:

$$\left| \frac{n+2}{3n+5} - L \right| = \underline{\hspace{2cm}}.$$

Now make the denominator smaller to get something simpler. (What is $3n+5$ bigger than? You may need to handle an extra constant factor in the denominator too.)

$$\underline{\hspace{2cm}} < \underline{\hspace{2cm}}.$$

Solve “simple bound $< \varepsilon$ ” for n :

$$n > \underline{\hspace{2cm}}.$$

3. The proof. Fill in the template:

We claim the limit is $\underline{\hspace{2cm}}$. Let $\varepsilon > 0$. By the Archimedean property, choose $N \in \mathbb{N}$ with $N > \underline{\hspace{2cm}}$. Then for all $n \geq N$,

$$\left| \frac{n+2}{3n+5} - \underline{\hspace{2cm}} \right| = \underline{\hspace{2cm}} < \underline{\hspace{2cm}} \leq \underline{\hspace{2cm}} < \varepsilon.$$

$$\text{Hence } \lim_{n \rightarrow \infty} \frac{n+2}{3n+5} = \underline{\hspace{2cm}}.$$

Check yourself. Can you justify each “=”, “<”, “ $\leq$ ” in your chain the same way the worked example does?